ONLY POINT MASSES ARE FREELY STRONGLY UNIMODAL

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Abstract. A probability measure μ on $\mathbb{R}$ is called freely strongly unimodal if $\mu \boxplus \nu$ is unimodal for every unimodal probability measure ν. Hasebe and Ueda showed that a measure with finite variance has this property only if it is a point mass, and asked whether any measure other than a point mass has it. We show that none does. For every probability measure μ that is not a point mass, the free convolution of μ with $\frac{1}{100}\delta_0 + \frac{99}{100}\,U[0,L]$, where $U[0,L]$ is the uniform law on $[0,L]$, fails to be unimodal for all sufficiently large L. No moment condition is needed. The proof works with a small dilation of μ, describes its inverse reciprocal Cauchy transform on a disc in the lower half-plane, and controls the resulting perturbation of the density by the Schwarz–Pick lemma.

1. Introduction

A Borel probability measure ν on $\mathbb{R}$ is *unimodal* if there is $a \in \mathbb{R}$ such that

$$\nu = \nu(\{a\})\,\delta_a + f(x)\,dx,$$

where f is nondecreasing on $(-\infty, a)$ and nonincreasing on (a, ∞). In classical probability, μ is strongly unimodal if $\mu * \nu$ is unimodal for every unimodal ν; by Ibragimov's theorem these are exactly the measures that are absolutely continuous, supported on an interval, with a log-concave density (or point masses), so the normal laws are examples [5, Theorem 52.3]. Hasebe and Ueda [4] introduced the free analogue.

Definition 1.1 ([4, Definition 3.8])**.** A probability measure μ on $\mathbb{R}$ is *freely strongly unimodal* if $\mu \boxplus \nu$ is unimodal for all unimodal distributions ν on $\mathbb{R}$.

Here $\boxplus$ is the free additive convolution, defined for arbitrary Borel probability measures by Bercovici and Voiculescu [2]. A point mass δ_a is freely strongly unimodal, since $\delta_a \boxplus \nu$ is the translate of ν by a. Hasebe and Ueda proved that semicircle laws are not freely strongly unimodal [4, Lemma 3.9], and deduced from the free central limit theorem that no measure with finite variance other than a point mass is [4, Theorem 3.10]. They asked:

> *Problem 3.11 of* [4]. Does there exist a freely strongly unimodal probability measure that is not a delta measure?

The answer is no.

Theorem 1.2. *Let μ be a Borel probability measure on $\mathbb{R}$ that is not a point mass, and for $L > 0$ let*

$$\nu_L = \tfrac{1}{100}\,\delta_0 + \tfrac{99}{100}\,U[0,L].$$

Each ν_L is unimodal, and there is $L_0 = L_0(\mu)$ such that $\mu \boxplus \nu_L$ is not unimodal for every $L > L_0$. Consequently the freely strongly unimodal probability measures on $\mathbb{R}$ are exactly the point masses.

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The finite-variance argument of [4] rescales $\mu^{\boxplus n}$ towards a semicircle law and so needs the second moment. Here we scale μ the other way. With $\varepsilon = 1/L$, the dilation D_ε (the push-forward under $x \mapsto \varepsilon x$) carries $\mu \boxplus \nu_L$ to $D_\varepsilon \mu \boxplus \kappa$, where

$$\kappa = p\,\delta_0 + q\,U[0,1], \qquad p = \tfrac{1}{100}, \quad q = \tfrac{99}{100}. \tag{1.1}$$

The law κ has an atom at its mode 0 and the constant density q on $(0,1)$. As $\varepsilon \to 0$ the measure $D_\varepsilon \mu$ tends to δ_0, and $D_\varepsilon \mu \boxplus \kappa$ is a small perturbation of κ. The point is that the perturbation of the density on $(0,1)$ is not flat: to first order it is governed by a single holomorphic function, with values in the lower half-plane, evaluated along the curve $x \mapsto G_\kappa(x + i0)$, and the Schwarz–Pick lemma forces this function to have comparable size at two points of the curve. The curvature of the curve near $x = 1/100$, coming from the atom, then produces a dip in the density that no tail behaviour of μ can cancel. Section 2 collects the analytic facts we use, Section 3 proves Theorem 1.2, and Section 4 reports a numerical check.

2. Preliminaries

For a Borel probability measure η on $\mathbb{R}$ and z in the upper half-plane $\mathbb{C}^+$ put

$$G_\eta(z) = \int_{\mathbb{R}} \frac{\eta(ds)}{z - s}, \qquad F_\eta(z) = \frac{1}{G_\eta(z)}, \qquad h_\eta(z) = F_\eta(z) - z.$$

The function F_η maps $\mathbb{C}^+$ to $\mathbb{C}^+$.

Lemma 2.1. *$\operatorname{Im} F_\eta(z) \geq \operatorname{Im} z$ for $z \in \mathbb{C}^+$, with strict inequality at every z when η is not a point mass.*

Proof. We have $\operatorname{Im} G_\eta(z) = -\operatorname{Im} z \int |z - s|^{-2}\,\eta(ds)$, so

$$\operatorname{Im} F_\eta(z) = \frac{-\operatorname{Im} G_\eta(z)}{|G_\eta(z)|^2} = \operatorname{Im} z \cdot \frac{\int |z-s|^{-2}\,\eta(ds)}{\left|\int (z-s)^{-1}\,\eta(ds)\right|^2}.$$

By the Cauchy–Schwarz inequality the last quotient is at least 1, with equality only if $s \mapsto (z-s)^{-1}$ is η-almost everywhere constant, that is, only if η is a point mass. □

We use the analytic subordination theorem for free additive convolution of arbitrary Borel probability measures [3, 1, 6]: if $\sigma = \lambda \boxplus \kappa$, there are analytic maps $\omega_\lambda, \omega_\kappa \colon \mathbb{C}^+ \to \mathbb{C}^+$ with

$$F_\lambda(\omega_\lambda(z)) = F_\kappa(\omega_\kappa(z)) = F_\sigma(z), \qquad \omega_\lambda(z) + \omega_\kappa(z) = z + F_\sigma(z). \tag{2.1}$$

We need that (2.1) determines $F_\sigma(z)$ pointwise.

Lemma 2.2. *Fix $z \in \mathbb{C}^+$. If $\omega_1, \omega_2, u \in \mathbb{C}^+$ satisfy $F_\lambda(\omega_1) = F_\kappa(\omega_2) = u$ and $\omega_1 + \omega_2 = z + u$, then $u = F_\sigma(z)$.*

Proof. From the two equations, $\omega_1 = z + h_\kappa(\omega_2)$ and $\omega_2 = z + h_\lambda(\omega_1)$, so ω_2 is a fixed point of $T_z(v) = z + h_\lambda(z + h_\kappa(v))$. By Lemma 2.1, h_κ and h_λ map $\mathbb{C}^+$ into $\{\operatorname{Im} \geq 0\}$, so T_z is a holomorphic map of $\mathbb{C}^+$ into $\{\operatorname{Im} v \geq \operatorname{Im} z\}$; in particular it is not the identity. A holomorphic self-map of $\mathbb{C}^+$ other than the identity has at most one fixed point in $\mathbb{C}^+$, by the Schwarz lemma transported to the half-plane. The subordination functions provide one fixed point, $\omega_\kappa(z)$, so $\omega_2 = \omega_\kappa(z)$ and $u = F_\kappa(\omega_2) = F_\sigma(z)$. □

Lemma 2.3. *Let μ be a probability measure that is not a point mass, $\lambda_\varepsilon = D_\varepsilon \mu$, and let $\mathcal{D}$ be an open disc whose closure lies in the lower half-plane. Put $U = \{1/w : w \in \overline{\mathcal{D}}\}$, a compact subset of $\mathbb{C}^+$. There are a bounded open set V with $U \subset V \subset \overline{V} \subset \mathbb{C}^+$ and $\varepsilon_0 > 0$ such that for $0 < \varepsilon < \varepsilon_0$:*

(i) *F_{λ_ε} is injective on V, and there is a holomorphic H_ε on a neighbourhood of U with values in V and $F_{\lambda_\varepsilon} \circ H_\varepsilon = \mathrm{id}$;*
(ii) *the function $R_\varepsilon(w) = H_\varepsilon(1/w) - 1/w$ is holomorphic on a neighbourhood of $\overline{\mathcal{D}}$, satisfies $\operatorname{Im} R_\varepsilon < 0$ there, and $\sup_{\overline{\mathcal{D}}} |R_\varepsilon| \to 0$ as $\varepsilon \to 0$.*

Proof. For $z \in \mathbb{C}^+$, $G_{\lambda_\varepsilon}(z) = \int (z - \varepsilon s)^{-1} \mu(ds)$, and $|z - \varepsilon s| \geq \operatorname{Im} z$, so by dominated convergence $G_{\lambda_\varepsilon}(z) \to 1/z$ pointwise. The family is bounded by $1/\operatorname{Im} z$, so by Vitali's theorem the convergence is locally uniform on $\mathbb{C}^+$, together with all derivatives, and the same holds for $F_{\lambda_\varepsilon} \to \mathrm{id}$.

Let $\delta > 0$ be less than half the distance from U to $\mathbb{R}$, and let V be the open 2δ-neighbourhood of the convex hull of U; it is convex, bounded, and $\overline{V} \subset \mathbb{C}^+$. Choose ε_0 so that $|F'_{\lambda_\varepsilon} - 1| < 1/2$ and $|F_{\lambda_\varepsilon} - \mathrm{id}| < \delta/2$ on $\overline{V}$ for $\varepsilon < \varepsilon_0$. Integrating along segments gives $\operatorname{Re} \frac{F_{\lambda_\varepsilon}(\omega) - F_{\lambda_\varepsilon}(\omega')}{\omega - \omega'} > \frac{1}{2}$ on V, so F_{λ_ε} is injective there. For u' within $\delta/2$ of U, Rouché's theorem on the circle $|\omega - u'| = \delta$ (which lies in V) shows that $F_{\lambda_\varepsilon}(\omega) = u'$ has exactly one solution in that disc; this defines H_ε, which is holomorphic as the inverse of an injective holomorphic map. Since $H_\varepsilon(u') - u' = H_\varepsilon(u') - F_{\lambda_\varepsilon}(H_\varepsilon(u'))$, we get $|H_\varepsilon(u') - u'| \leq \sup_V |F_{\lambda_\varepsilon} - \mathrm{id}| \to 0$.

This gives (i) and the last assertion of (ii). For the sign, let $u = 1/w$ with w near $\overline{\mathcal{D}}$ and $\omega = H_\varepsilon(u) \in \mathbb{C}^+$. Since λ_ε is not a point mass, Lemma 2.1 gives $\operatorname{Im} u = \operatorname{Im} F_{\lambda_\varepsilon}(\omega) > \operatorname{Im} \omega$, so $\operatorname{Im} R_\varepsilon(w) = \operatorname{Im}(\omega - u) < 0$. □

No moment of μ enters Lemma 2.3, and no global inverse of F_{λ_ε} is used.

3. Proof of Theorem 1.2

3.1. **The reference measure and the disc.** Let κ be as in (1.1). Then

$$G_\kappa(z) = \frac{p}{z} + q\big(\log z - \log(z-1)\big),$$

and for $0 < x < 1$ its boundary value from $\mathbb{C}^+$ is $G_\kappa(x + i0) = g(x) - ic$, where

$$g(x) = \frac{p}{x} + q \log \frac{x}{1-x}, \qquad c = \pi q.$$

For z near a point of $(0,1)$ put

$$A(z) = \frac{p}{z} + q\big(\operatorname{Log} z - \operatorname{Log}(1-z) - i\pi\big)$$

with the principal branch of Log. Then A is holomorphic on a neighbourhood of $(0,1)$, agrees with G_κ on its intersection with $\mathbb{C}^+$, and $A(x) = g(x) - ic$ for real $x \in (0,1)$. In particular $A'(x) = g'(x)$ is real there. Direct differentiation gives

$$g'(x) = \frac{x - p}{x^2(1-x)}, \qquad g''(x) = \frac{2p}{x^3} + q\Big(\frac{1}{(1-x)^2} - \frac{1}{x^2}\Big). \tag{3.1}$$

Let $x_1 = p = 1/100$ and $x_2 = 1/4$. By (3.1),

$$g'(x_1) = 0, \qquad g''(x_1) = \frac{1000000}{99}, \qquad g'(x_2) = \frac{128}{25}, \qquad g''(x_2) = -\frac{64}{5}. \tag{3.2}$$

Put $w_j = g(x_j) - ic$ and

$$b = g(x_2) - 10i, \qquad r = \tfrac{99}{10}, \qquad \mathcal{D} = \{w : |w - b| < r\}.$$

The closure of $\mathcal{D}$ lies in $\{\operatorname{Im} w \leq -1/10\}$. Since $3 < \pi < 4$ we have $2.97 < c < 3.96$. Let $d = g(x_1) - g(x_2) = \frac{24}{25} - \frac{99}{100} \log 33$; since $1 < \log 33 < 4$, $|d| < 3$. Now $w_2 - b = i(10 - c)$ and $w_1 - b = d + i(10 - c)$, so

$$|w_2 - b| < 7.03 < r, \qquad |w_1 - b|^2 < 9 + 7.03^2 < r^2,$$

and both w_j lie in $\mathcal{D}$. Write $\Delta = r^2 - (10-c)^2$. Then

$$\Delta > 98.01 - 7.03^2 = 48.5891, \qquad \frac{2r}{\Delta} < \frac{19.8}{48.5891} < \frac{5}{12}.$$

The pseudohyperbolic distance of w_1 and w_2 in $\mathcal{D}$ is

$$\rho = \frac{r|w_1 - w_2|}{|r^2 - (w_1 - b)\overline{(w_2 - b)}|} = \frac{r|d|}{|\Delta + id(10-c)|} \le \frac{r|d|}{\Delta} < \frac{29.7}{48.5891} < \frac{2}{3}.$$

(Numerically $\Delta \approx 50.54$, $d \approx -2.5015$ and $\rho \approx 0.464$.)

3.2. **The perturbation and its size.** Fix μ, not a point mass, and apply Lemma 2.3 to $\lambda_\varepsilon = D_\varepsilon \mu$ and the disc $\mathcal{D}$. Put

$$a_\varepsilon = \operatorname{Re} R_\varepsilon(w_2), \qquad t_\varepsilon = -\operatorname{Im} R_\varepsilon(w_2) > 0, \qquad B_\varepsilon = R_\varepsilon - a_\varepsilon.$$

Then $t_\varepsilon \to 0$, B_ε maps $\mathcal{D}$ into the lower half-plane, and $B_\varepsilon(w_2) = -it_\varepsilon$. Write $t = t_\varepsilon$, $B = B_\varepsilon$ and $K = -\operatorname{Im} B > 0$ on $\mathcal{D}$.

Lemma 3.1. *For every compact $M \subset \mathcal{D}$ and every $k \ge 0$ there is $C_{M,k}$, independent of ε, with $|B^{(k)}| \le C_{M,k}\, t$ on M. Moreover*

$$K(w_1) > \tfrac{1}{5}\, t, \qquad K(w_2) = t, \qquad |\partial_x K(w_2)| < \tfrac{5}{12}\, t,$$

where ∂_x is the derivative in the real direction.

Proof. Since $\operatorname{Im} B < 0$ and $t > 0$, the function $\phi = (B+it)/(B-it)$ maps $\mathcal{D}$ into the unit disc and $\phi(w_2) = 0$. By the Schwarz–Pick lemma, $|\phi(w)| \le \rho_{\mathcal{D}}(w, w_2)$ and $|\phi'(w_2)| \le r/(r^2 - |w_2 - b|^2) = r/\Delta$. Solving for B,

$$B = -it\,\frac{1+\phi}{1-\phi}, \qquad K = t\,\frac{1-|\phi|^2}{|1-\phi|^2} \;\ge\; t\,\frac{1-|\phi|}{1+|\phi|}.$$

On a compact $M \subset \mathcal{D}$ we have $\sup_M |\phi| \le \sup_{w \in M} \rho_{\mathcal{D}}(w, w_2) < 1$, which bounds B by a multiple of t on M; Cauchy's estimates on a slightly larger compact set bound the derivatives. At w_1, $K(w_1) \ge t(1-\rho)/(1+\rho) > t \cdot \frac{1/3}{5/3} = t/5$. Finally $B'(w_2) = -2it\,\phi'(w_2)$, so $|\partial_x K(w_2)| = |\operatorname{Im} B'(w_2)| \le |B'(w_2)| \le 2rt/\Delta < \frac{5}{12} t$. □

3.3. **The density of the convolution.** Let $\tilde{\lambda}_\varepsilon$ be the translate of λ_ε by $-a_\varepsilon$, so that $F_{\tilde{\lambda}_\varepsilon}(\omega) = F_{\lambda_\varepsilon}(\omega + a_\varepsilon)$, and let $\sigma_\varepsilon = \tilde{\lambda}_\varepsilon \boxplus \kappa$. Fix $j \in \{1, 2\}$ and a closed disc N_j centred at x_j, small enough that A is holomorphic on a neighbourhood of N_j and $A(N_j)$ is contained in a compact set $M \subset \mathcal{D}$ with $\operatorname{dist}(A(N_j), \partial M) > 0$.

For $z \in N_j$ consider the equation

$$W = A(z - B(W)). \tag{3.3}$$

By Lemma 3.1, B and B' are $O(t)$ on M, so for small ε the right side of (3.3) maps the closed disc of radius Ct about $A(z)$ into itself and is a contraction there, for a suitable C independent of ε and z, after shrinking N_j slightly. The fixed point $W_\varepsilon(z)$ is holomorphic in z (the iterates converge uniformly), lies in M, and satisfies $W_\varepsilon = A + O(t)$ uniformly on N_j.

Lemma 3.2. *$W_\varepsilon(z) = G_{\sigma_\varepsilon}(z)$ for $z \in N_j \cap \mathbb{C}^+$.*

Proof. Let $z \in N_j \cap \mathbb{C}^+$, $W = W_\varepsilon(z)$ and $u = 1/W \in U$. Put

$$\omega_1 = H_\varepsilon(u) - a_\varepsilon = u + B(W), \qquad \omega_2 = z - B(W).$$

Then $\omega_1 \in \mathbb{C}^+$ because $H_\varepsilon(u) \in V$, and $\operatorname{Im} \omega_2 = \operatorname{Im} z + K(W) > 0$. By Lemma 2.3, $F_{\tilde{\lambda}_\varepsilon}(\omega_1) = F_{\lambda_\varepsilon}(H_\varepsilon(u)) = u$. Since $\omega_2 \in \mathbb{C}^+$ is near x_j, $A(\omega_2) = G_\kappa(\omega_2)$, so (3.3) says $F_\kappa(\omega_2) = 1/W = u$. Finally $\omega_1 + \omega_2 = z + u$. Lemma 2.2 gives $u = F_{\sigma_\varepsilon}(z)$, that is $W = G_{\sigma_\varepsilon}(z)$. □

Thus G_{σ_ε} extends holomorphically across the real interval $N_j \cap \mathbb{R}$, and by Stieltjes inversion σ_ε is absolutely continuous there with the real-analytic density

$$f_\varepsilon(x) = -\frac{1}{\pi} \operatorname{Im} W_\varepsilon(x), \qquad x \in N_j \cap \mathbb{R}.$$

3.4. **First-order expansion.** Expanding (3.3) with $W_\varepsilon = A + O(t)$ and $B, B' = O(t)$ on M,

$$W_\varepsilon(z) = A(z) - A'(z)\, B(W_\varepsilon(z)) + O(t^2) = A(z) - A'(z)\, B(A(z)) + O(t^2)$$

uniformly on N_j, with constants independent of ε. The remainder is holomorphic, so by Cauchy's estimates its derivative is also $O(t^2)$ on a smaller concentric disc. For real x we have $A(x) = g(x) - ic$ and $A'(x) = g'(x) \in \mathbb{R}$, hence

$$f_\varepsilon(x) = q - \frac{1}{\pi}\, g'(x)\, K\big(A(x)\big) + O(t^2).$$

As x moves along the real axis, $A(x)$ moves horizontally with speed $g'(x)$, so $\frac{d}{dx} K(A(x)) = g'(x)\, \partial_x K(A(x))$ and

$$f'_\varepsilon(x) = -\frac{1}{\pi}\Big(g''(x)\, K\big(A(x)\big) + g'(x)^2\, \partial_x K\big(A(x)\big)\Big) + O(t^2). \tag{3.4}$$

3.5. **Conclusion.** At x_1, (3.2), (3.4) and Lemma 3.1 give

$$f'_\varepsilon(x_1) = -\frac{g''(x_1)}{\pi} K(w_1) + O(t^2) \le -\frac{200000}{99\,\pi}\, t + O(t^2).$$

At x_2,

$$f'_\varepsilon(x_2) \ge \frac{1}{\pi}\Big(\frac{64}{5} - \Big(\frac{128}{25}\Big)^2 \frac{5}{12}\Big) t - O(t^2) = \frac{704}{375\,\pi}\, t - O(t^2).$$

Since $t = t_\varepsilon > 0$ and $t_\varepsilon \to 0$, there is $\varepsilon_1 > 0$ such that for $0 < \varepsilon < \varepsilon_1$ the density f_ε of σ_ε is strictly decreasing near x_1 and strictly increasing near x_2. Then σ_ε is not unimodal: if a were a mode, the density would be nondecreasing almost everywhere to the left of a, which forces $a \le x_1$, and nonincreasing to the right of a, which fails near $x_2 > x_1$.

Finally, $D_\varepsilon(\mu \boxplus \nu_{1/\varepsilon}) = D_\varepsilon \mu \boxplus D_\varepsilon \nu_{1/\varepsilon} = \lambda_\varepsilon \boxplus \kappa$, and σ_ε is the translate of this measure by $-a_\varepsilon$. Translations and positive dilations preserve unimodality, so $\mu \boxplus \nu_L$ is not unimodal for $L > 1/\varepsilon_1$. Each ν_L is unimodal with mode 0, so μ is not freely strongly unimodal. Together with the remark on point masses in the introduction this proves Theorem 1.2. □

Remark 3.3. The atom of κ is what makes g' vanish at an interior point of the support, with g'' large there. The proof gives no explicit $L_0(\mu)$, since t_ε is not quantified in terms of μ.

4. A numerical check

As an independent check of the sign pattern we computed the density of $\sigma \boxplus \kappa$ directly for small σ of two kinds. For the semicircle law of variance v the subordination equation reduces to $G(z) = G_\kappa(z - vG(z))$, solved by fixed-point iteration at $z = x + 10^{-13} i$; for the symmetric Bernoulli law on $\{\pm a\}$ we iterated $\omega \mapsto z + h(z + h_\kappa(\omega))$ with $h(w) = -a^2/w$. Central differences with step 10^{-5} gave the following.

law of σ	$f'(0.01)$	$f'(0.25)$	interior local minimum
semicircle, variance 10^{-6}	-9.99×10^{-3}	1.27×10^{-5}	$x \approx 0.020$
semicircle, variance 10^{-4}	-8.66×10^{-1}	1.26×10^{-3}	$x \approx 0.020$
Bernoulli $\pm 10^{-3}$	-9.99×10^{-3}	1.27×10^{-5}	$x \approx 0.020$

In each case the density has an interior local minimum just to the right of x_1, as the proof predicts. These measures have finite variance, so they are covered by [4, Theorem 3.10]; the computation only confirms the mechanism. The arithmetic of Section 3 (the values (3.2), the bounds on Δ, ρ and $2r/\Delta$, and the two slopes) was also checked in exact rational and 50-digit arithmetic.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In October 2026 a GPT-family model (OpenAI), running in the Codex command-line tool, found the proof of Theorem 1.2; a Claude model (Anthropic) reviewed it. In November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, checked the proof and its constants, wrote and ran the computations of Section 4, and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code for the computations of Section 4 and for the check of the constants, with their outputs, is available at
`https://github.com/yspennstate/free-strong-unimodality-point-masses`.

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